

Inyoung Ryu

Texas A&M University
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RESEARCH INTERESTS

- Surface group representations and character varieties
- Mapping class groups and Teichmüller theory
- Low-dimensional topology and geometric group theory

EDUCATION

Texas A&M University <i>Ph.D. in Mathematics. Expected graduation date: May 2027</i>	Aug. 2021 – Present Texas, US
Ulsan National Institute of Science and Technology (UNIST) <i>Bachelor of Science in Mathematical Sciences</i> • Minor in Computer Science and Engineering	Mar. 2017 – Jul. 2021 Ulsan, Korea
Sabancı University <i>Erasmus Program</i>	Feb. 2020 – Jun. 2020 Istanbul, Turkey

PUBLICATIONS AND PREPRINTS

From Bowditch's question to Goldman's conjecture for type-preserving representations <i>with Viraj Joshi, submitted, arXiv: 2609.02065</i>	Sep. 2026
On totally hyperbolic non-Fuchsian type-preserving representations <i>Submitted, arXiv: 2511.13989</i>	Nov. 2025
Connected components of the space of type-preserving representations <i>with Tian Yang, submitted, arXiv: 2507.07391</i>	Jul. 2025

INVITED SEMINARS AND COLLOQUIUM TALKS

(Upcoming) Two invariants of type-preserving representations and applications <i>Joint Mathematics Meetings (JMM 2027)</i>	Jan. 2027 Illinois, US
(Upcoming) Two invariants of type-preserving representations and applications <i>Geometric Group Theory Seminar, Purdue University</i>	Nov. 2026 Indiana, US
Two invariants of type-preserving representations and applications <i>Geometry and Dynamics Seminar, Centro de Ciencias Matemáticas UNAM (Online)</i>	Sep. 2026 Morelia, Mexico
Two invariants of type-preserving representations and applications <i>Topology Seminar, University of Tennessee</i>	Apr. 2026 Tennessee, US
Two invariants of type-preserving representations and applications <i>Topology & Geometry Seminar, Rutgers University</i>	Apr. 2026 New Jersey, US
Two invariants of type-preserving representations and applications <i>Algebra & Combinatorics Seminar, Texas A&M University</i>	Mar. 2026 Texas, US
Two invariants of type-preserving representations and applications <i>Groups & Dynamics Seminar, University of Texas Austin</i>	Mar. 2026 Texas, US
Two invariants of type-preserving representations and applications <i>Groups & Dynamics Seminar, Texas A&M University</i>	Feb. 2026 Texas, US
Connected components of the space of type-preserving representations <i>SHSU Junior Colloquium, Sam Houston State University</i>	Nov. 2025 Texas, US
Connected components of the space of type-preserving representations <i>KIAS HCMC Topology Seminar, Korea Institute for Advanced Study (Online)</i>	Feb. 2025 Seoul, Korea
Geometric approach to surface group representations <i>Graduate Student Organization Seminar, Texas A&M University</i>	Sep. 2024 Texas, US

Topological components of spaces of representations*SNU Topology Seminar, Seoul National University (Online)*

Aug. 2024

Seoul, Korea

Mapping class group action on Teichmüller Space*Graduate Student Organization Seminar, Texas A&M University*

Apr. 2024

Texas, US

CONTRIBUTED TALKS

(Upcoming) On totally hyperbolic non-Fuchsian type-preserving representations*AMS 2026 Fall Western Sectional Meeting, Arizona State University*

Nov. 2026

Arizona, US

(Upcoming) Non-Fuchsian totally hyperbolic type-preserving representations*AMS 2026 Fall Central Sectional Meeting, University of Minnesota Duluth*

Oct. 2026

Minnesota, US

Connected components of the space of type-preserving representations*GTA: Philadelphia 2025, Temple University*

May 2025

Pennsylvania, US

Connected components of the space of type-preserving representations*CIRM School on Geometric structures and discrete group actions*

Apr. 2025

Marseille, France

Connected components of the space of type-preserving representations*Graduate Student Conference in Topology and Geometry, Indiana University Bloomington*

Apr. 2025

Indiana, US

Nielsen-Thurston classification*GIGEM Conference, Texas A&M University*

Apr. 2024

Texas, US

LIGHTNING TALKS AND POSTER PRESENTATIONS

Can a non-Fuchsian representation be totally hyperbolic?*Vanderbilt colloquium series: Groups, geometry, and dynamics (ICM satellite conference)*

Jul. 2026

Tennessee, US

Poster Presentation: Non-Fuchsian totally hyperbolic representations*Young Geometric Group Theory XIV*

Jun. 2026

Seoul, Korea

Connected components of the space of type-preserving representations*Ventotene Workshop on Higher-dimensional hyperbolic geometry*

Sep. 2025

Ventotene, Italy

Connected components of the space of type-preserving representations*2K-GATE Workshop, Korea Institute for Advanced Study*

Jun. 2025

Seoul, Korea

Connected components of the space of type-preserving representations*53rd Barrett Lectures: Recent Development in Geometric Group Theory*

May 2025

Tennessee, US

Connected components of the space of type-preserving representations*Conference on Geometry and Dynamics in Low Dimensions, IMVAST*

Jan. 2025

Hanoi, Vietnam

SELECTED CONFERENCES AND WORKSHOPS

International Congress of Mathematicians (ICM 2026)

Jul. 2026

Pennsylvania, US

Joint Mathematics Meetings (JMM 2026)

Jan. 2026

Washington, D.C., US

Workshop on Homogeneous Dynamics and Anosov Representations*Simons Laufer Mathematical Sciences Institute*

Apr. 2026

California, US

SLMath Summer School: Geometry and Dynamics in Higher Rank Lie Groups*University of California, Berkeley*

Jul. 2025

California, US

Workshop on Low-dimensional phenomena: geometry and dynamics*Institut Henri Poincaré*

Jun. 2025

Paris, France

Workshop on Lorentzian, Affine, and Hyperbolic Differential Geometry*University of Maryland*

Apr. 2025

Maryland, US

Workshop on Motives and Mapping Class Groups*American Institute of Mathematics, Caltech*

Jan. 2025

California, US

Joint Mathematics Meetings (JMM 2024)

Jan. 2024

California, US

Conference on Characters and Moduli of Surfaces*Kyoto University*

Jul. 2023

Kyoto, Japan

HONORS, AWARDS, AND GRANTS

AMS Fall 2026 Sectional Meeting Travel Grant <i>American Mathematical Society</i>	Sep. 2026 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang <i>Texas A&M University</i>	Sep. 2026 - Dec. 2026 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang <i>Texas A&M University</i>	Jul. 2024 – Dec. 2024 <i>Texas, US</i>
Houston A&M Mothers' Club Outstanding TA Award <i>Awarded for going above and beyond expectations as a teaching assistant</i>	Jul. 2024 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, PI: Tian Yang <i>Texas A&M University</i>	Jan. 2024 – May 2024 <i>Texas, US</i>

TEACHING

Python Lab Assistant, MATH 151 & 152 (Engineering Mathematics I & II) <i>Texas A&M University. Creating Python lab assignments.</i>	Jul. 2026 <i>Texas, US</i>
Instructor of record, MATH 142 (Business Calculus) <i>Texas A&M University. Lecturing, holding office hours. Writing, proctoring, and grading exams.</i>	Jan. 2026 - May 2026 <i>Texas, US</i>
Grader, MATH 636 (Topology I) <i>Texas A&M University</i>	Sep. 2025 - Dec. 2025 <i>Texas, US</i>
Python Lab Assistant, MATH 150 (Functions, Trigonometry and Linear Systems) <i>Texas A&M University. Creating Python lab assignments.</i>	Jul. 2025 <i>Texas, US</i>
Grader, MATH 467 (Modern Geometry) <i>Texas A&M University</i>	Jan. 2025 - May 2025 <i>Texas, US</i>
Help Session Tutor, MATH 142 (Business Calculus) <i>Texas A&M University</i>	Jun. 2024 <i>Texas, US</i>
Grader, MATH 636 (Topology I) <i>Texas A&M University</i>	Aug. 2023 – Dec. 2023 <i>Texas, US</i>
Grader, MATH 142 (Business Calculus) <i>Texas A&M University</i>	Jul. 2023 <i>Texas, US</i>
Python Lab Assistant, MATH 151 (Engineering Math I) <i>Texas A&M University. Creating Python lab assignments.</i>	Jun. 2023 <i>Texas, US</i>
Teaching Assistant, MATH 152 (Engineering Math II) <i>Texas A&M University. Leading recitations.</i>	Jan. 2023 – May 2023 <i>Texas, US</i>
Teaching Assistant, MATH 151 (Engineering Math I) <i>Texas A&M University. Leading recitations.</i>	Aug. 2022 – Dec. 2022 <i>Texas, US</i>
Grader, MATH 304 (Linear Algebra) <i>Texas A&M University</i>	Jun. 2022 <i>Texas, US</i>
Grader, MATH 423 (Linear Algebra II) <i>Texas A&M University</i>	Jan. 2022 – May 2022 <i>Texas, US</i>
Grader, MATH 467 (Modern Geometry) <i>Texas A&M University</i>	Aug. 2021 – Dec. 2021 <i>Texas, US</i>
Teaching Assistant (Calculus I, Calculus II) <i>Ulsan National Institute of Science and Technology. Assistant in discussion sessions</i>	Mar. 2018 – Dec. 2019 <i>Ulsan, Korea</i>
Calculus I Open Lecture <i>Ulsan National Institute of Science and Technology</i>	Jun. 2018 <i>Ulsan, Korea</i>

MENTORING

Directed Reading Program: A first course in Algebraic Topology <i>Texas A&M University</i>	Sep. 2025 - Dec. 2025 <i>Texas, US</i>
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SERVICE

Treasurer, Association for Women in Mathematics
Texas A&M University

Aug. 2022 - May 2023
Texas, US

- Women in Math Mentoring Lunch
- AWM Peer Mentoring Program

Teaching Assistant for Math Circle
Texas A&M Math Circle

Nov. 2023
Texas, US

Grader for High School Math Contest
Texas A&M High School Mathematics Contest

Nov. 2023
Texas, US

Grader for High School Math Contest
Texas A&M High School Mathematics Contest

Nov. 2022
Texas, US